



Hydraulic & Engineering Instruments

# Progressive in Design Perfect in Performance

## OPERATING MANUAL



An ISO 9001 : 2015 Company



**OPERATING INSTRUCTIONS  
FOR  
HC 37.65 VEE-BEE CONSISTOMETER**

**Hydraulic & Engineering Instruments  
(An ISO 9001:2015 Company)**

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**ISO 9001:2015**

## **C O N T E N T S**

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**SETTING UP & TEST PROCEDURE**

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### **EQUIPMENT**

The Equipment Comprises the following:-

1. One vibrator with rubber supports.
2. One No. Slump Cone.
3. One No. Metal Pot
4. One swivel arm with funnel
5. One graduated rod with perspex disc and annular weight
6. One tamping rod
7. Instruction Manual with G. A Drawing

## INTRODUCTION

This apparatus meets the essential requirements of IS:1199-1959 and IS: 10510-1983 for the determining of consistency of concrete by determining time required for transforming by vibration, a concrete specimen in the form of a conical frustum into a cylinder.

## DESCRIPTION

The apparatus comprises a vibrating table supported on rubber shock-absorbers on a base (1). A 3 phase electric motor with eccentric masses on both sides of its shaft, is fixed underneath the vibrating table. A switch for the operation of the motor and a cable for connecting to 440volt, 3 phase mains are fitted on the base (1). A swivel arm holder (5) is also attached to the base. A cylindrical metal container (3) can be fixed at the top of vibrating table. A slump cone (2) is provided for the preparation of conical frustum of concrete specimen.

A swivel arm (6) telescoped into the swivel arm holder (5) carries a funnel (4) on one side and a guide sleeve (7) on the other side. A graduated rod passes through the guide sleeve & carries a transparent perspex disc with an annular weight at its lower end. This rod can be locked at any desired position with a knurled screw in the guide sleeve. A tamping rod is provided for tamping during specimen preparation. The apparatus is suitable for operation on 440volts, 3 phase, 50 cycles, A/C supply.

## SETTING UP & TEST PROCEDURE

Fit the metal container on the top of vibrating table and place the metal cone in the metal container. Fit the swivel arm (6) in the swivel arm holder (5). Connect the apparatus to the main supply i.e.440volts, 3phase, A/C supply.

Swing the swivel arm (6) and place the funnel (4) on the top of the slump cone (2), place the concrete specimen in the metal cone in four equal layers tamping each layer with 25 well distributed strokes of the tamping rod. While compacting second and subsequent layers the strokes should be so given that they just penetrate into the underlying layer. After top layer has been rodded, the funnel (4) is swung away from the metal cone taking care that the excess concrete does not fall into the metal cylinder. Strike off the excess material from the top of the cone to level with its top.

Swing the swivel arm so that the axis of graduated rod is on the top of specimen. Gently place the disc (8) on the top of slump cone and take the initial reading on the graduated rod. Swing the transparent disc away clear of the cone and metal container. Raise the metal cone slowly and carefully in a vertical measure the slump by carefully placing the perspex disc (8) on the subsided specimen.

Then switch on the electrical vibrator and allow the concrete to spread out in the metal pot. Continue the vibration until the whole concrete surface uniformly touches the bottom surface of the perspex disc. The time taken to attain this should be noted in seconds with a stop watch.

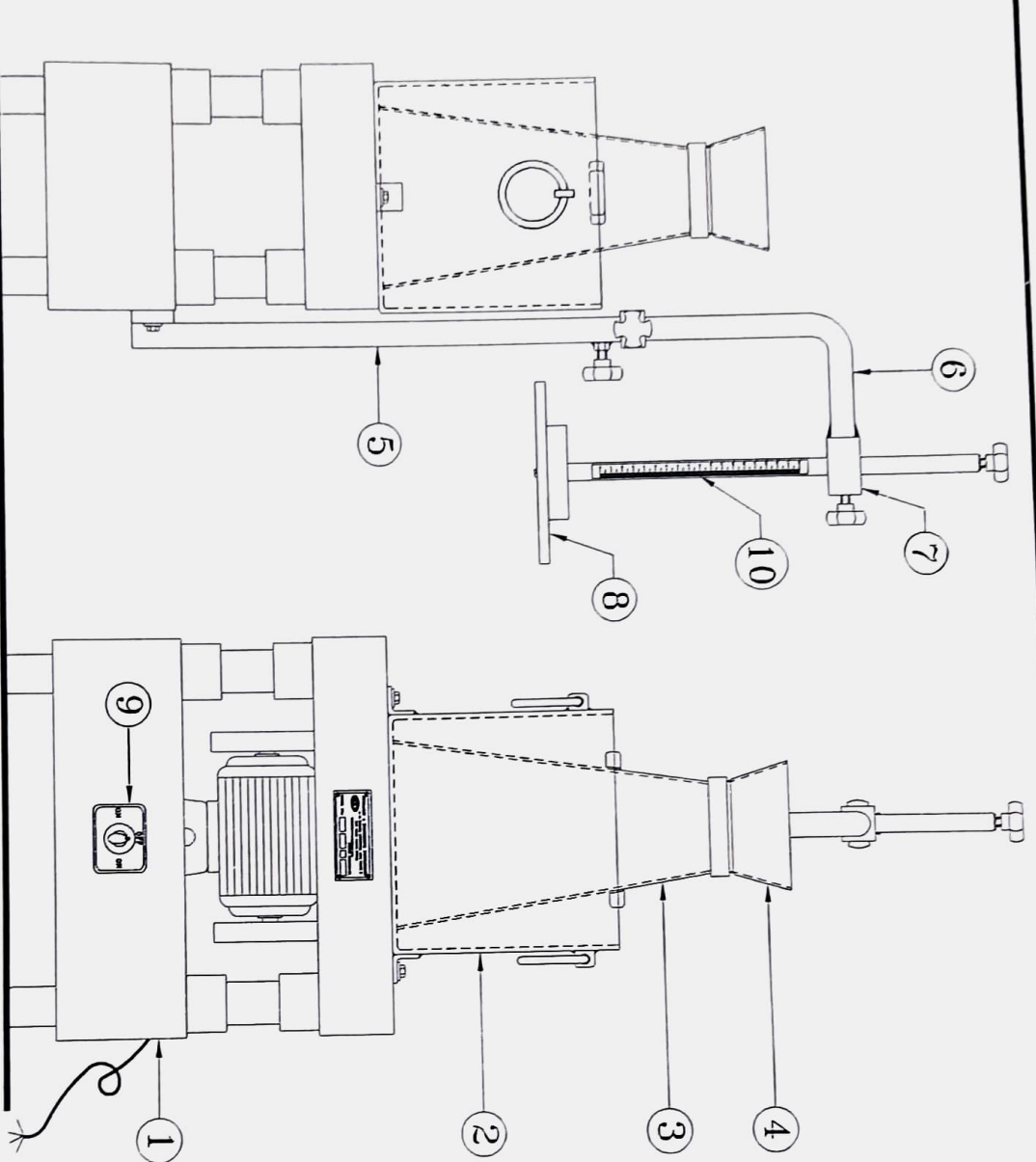
The consistency of concrete shall be expressed in VB-number which is equal to the time taken in seconds recorded as described above for the concrete to be transformed to a cylinder.

## MAINTENANCE

After the test, clean the metal pot, metal cone funnel and perspex disc free from any concrete particles sticking to them.

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| S. NO. | D E S C R I P T I O N |
|--------|-----------------------|
| 10     | SCALE                 |
| 09     | STARTER               |
| 08     | ACRALIC SHEET         |
| 07     | GUIDE SLEEVE          |
| 06     | SWIVEL ARM            |
| 05     | SWIVEL ARM HOLDER     |
| 04     | FUNNEL                |
| 03     | METAL CONE            |
| 02     | METAL POT             |
| 01     | BASE                  |



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NOTES:- GENERAL TOLERANCES FOR ALL DIMENSIONS NOT SPECIFIED & ALL DIMENSIONS IN MILLIMETER.

MACHINE:-VEE-BEE CONSISTOMETER

PART:-G.A. Drawing

|                    |             |
|--------------------|-------------|
| CAT.NO.<br>HCS7/65 | DRG.NO.     |
| QTY:-              | REV.NO:-    |
| DRN KAMAL          | CHD.H.SAHU  |
| APPD.              | SCALE : NTS |
| 08.09.2012         |             |